Probabilistic Design and Safety-Constrained Optimization of a Firewater-Powered Portable Valve Actuation System

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Abstract

Almost every hazardous facility owns a pressurized, regularly inspected source of hydraulic power: its firewater system. This study asks whether a portable water-hydraulic operator driven through a fire hose can stroke seized or stiff valves without overloading the stem or taking the water that firefighting needs, and which design and operating rules make that likely. A Monte Carlo model with an explicit hydraulic circuit, motor leakage, a relief valve and the surge on stall is evaluated on 200,000 draws; every added mechanism can be switched off, recovering the earlier first-order model exactly. Under assumed input distributions a baseline tool (200 cc, 250:1, 60 gpm) completes the stroke in 75% of the draws but without exceeding the allowable stem torque in only 63%, because a median surge of 45 psi accompanies the stall; a stroke needs about 1,600 L of water. A hose-side accumulator and a relief chart set from the stem strength reduce the over-torque probability to 1% (motor speed at most 3,000 rpm) at a cost of 3 to 9 percentage points of safe strokes, of which a slow creep flow at the start and end of travel recovers 4 to 6. The surge model reproduces classical water-hammer results. The optimum holds out of sample but not under a stem 30% weaker than its data sheet (14% over-torque), and the

firefighting reserve is protected only if the spare flow of the source is known. The results are simulation outputs under assumed priors, not measurements.

Keywords

Water hydraulics; Valve actuation; Firewater; Monte Carlo simulation; Multi-objective optimization; Emergency response

1. Introduction

Large valves must sometimes be operated when plant power, instrument air and hydraulic supply are lost. Manual effort is limited, and improvised extensions expose the operator to injury [1]. The reliability of emergency valves is usually assessed by partial-stroke testing of the installed actuator [2], which does not cover an actuator that has no power. Portable operators powered by batteries, compressed air, gasoline or an oil-hydraulic pack exist, but a battery tool depends on a charged pack and is an ignition-source concern in classified areas, a pneumatic tool depends on plant air, and a hydraulic pack is heavy and needs oil. An energy company has recently sought a portable, preferably non-electric device that multiplies an operator's torque four- to six-fold for valves of 2 to 16 inches, deployable in remote, cold regions [3].

Almost every hazardous facility already owns a pressurized, regularly inspected source of hydraulic power: the firewater system. A hose at 100 psi and 150 gpm carries about 6.5 kW, far more than a portable valve tool needs. Water-hydraulic motors are an established technology [4–6], the pressure relief of water circuits has been studied [7], and the hydraulics of fire hoses [8] and hydrant supplies [9,10] are well characterised. Patents describe hydrant-mounted operating apparatus [11], hydrant-attached remote-controlled valves [12] and oscillatory actuators for fire-truck valves [13]. To the author's knowledge, in a search that was not exhaustive, none treats the question that a designer of a firewater-powered valve operator has to answer: with uncertain

supply, valve demand and torque-limiter setting, how often does the tool complete the stroke without exceeding the allowable stem torque and without taking water that firefighting needs, and which design and operating rules make that likely?

The three requirements conflict: a higher relief setting breaks stiff valves free but overloads stems, a higher flow limit shortens the stroke but draws water that a fire may need, and a lower gear ratio speeds the stroke and lowers the torque. That pressurized firewater can drive a valve operator follows from established hydraulic principles; the contribution that can be defended is a reproducible design and operating envelope that treats these requirements together and says how far it can be trusted. The paper formulates a probabilistic model with an explicit hydraulic circuit and the pressure surge on stall, in which each added mechanism can be switched off so that the earlier first-order model is recovered exactly; defines the safety criteria separately (success of the stroke, exceedance of the allowable stem torque, damage to the stem, violation of the firefighting reserve); poses a multi-objective design problem (motor, ratio, relief chart, flow limit and slow creep flows at the start and end of travel) under a limit on the over-torque probability; and tests the optimum out of sample, under deliberately wrong input distributions, and against a transient simulation. All input distributions are assumed priors for a concept-stage study.

2. System architecture and operating rule

The tool (Fig. 1) is a portable water-hydraulic torque operator. A hose connects a hydrant, tap, fire truck or portable pump to a Smart Power Coupling with a cleanable strainer, a flow limiter, a slow-opening shut-off valve and a pressure gauge. Water drives a gerotor, piston or vane motor of 150 to 400 cc per revolution [5,6]; a planetary reducer of 100:1 to 500:1 multiplies the torque; a universal interface with a reaction arm engages handwheels, operating nuts and part-turn valves in the manner of the standard actuator attachments [14]. An adjustable inlet

relief valve limits the pressure drop across the motor, and therefore the torque; a stall shows as a pressure rise. Options are a hose-side accumulator to absorb the surge on stall and an adjustable flow limiter that runs at a slow creep flow for the first turns of the stroke, to build torque gradually against a stuck valve, and for the last turns, to approach the stop slowly. A discharge hose routes the exhaust water to a drain. The baseline is purely mechanical, which keeps it clear of the electrical ignition concerns of classified areas, under the assessment of non-electrical equipment [15].

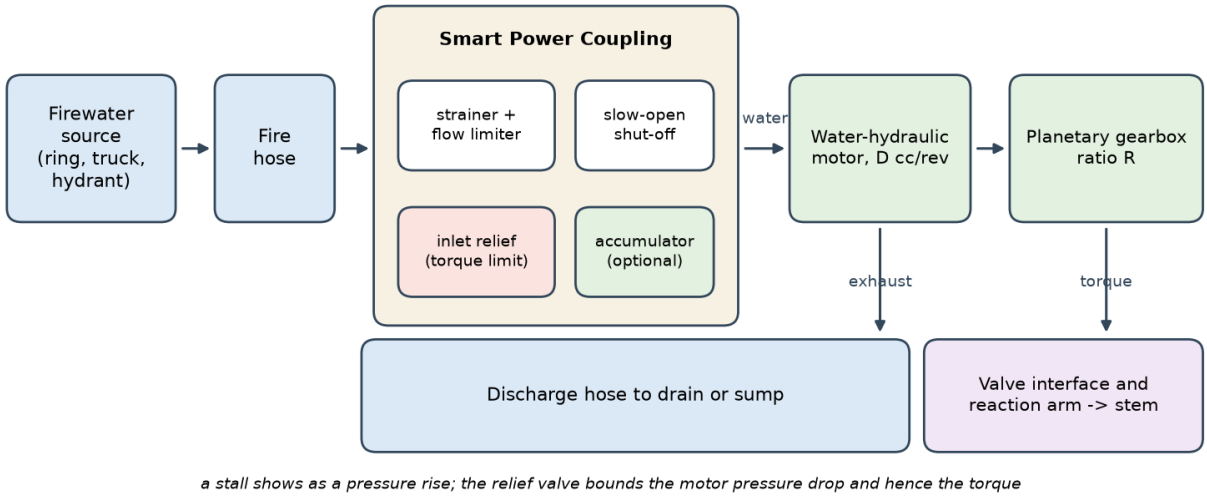


Fig. 1. Architecture of the firewater-powered valve operator. Water from a shared or dedicated source reaches the Smart Power Coupling (strainer and flow limiter, slow-opening shut-off, inlet relief valve that limits the torque, optional accumulator), drives the motor and gearbox, and is discharged through a hose. The torque is delivered to the valve stem through the interface and a reaction arm.

The displacement D and the gear ratio R are chosen from a kit; the relief setting, the flow limit and the creep flows are operating variables. The relief setting is read from a torque chart, the safety-critical document of the tool, on one of two bases: a multiple of the median breakaway

torque of the valve class, which is what a designer who knows only the operating torque would use, or a fraction of the nominal allowable stem torque, which requires the valve data sheet. The tool is subject to one operating rule that the model enforces: it must not draw water from a fire-protection system when doing so could compromise the performance that system must deliver. A shared source (a firewater ring or hydrant) may be used only up to the flow that it can spare without cutting its firefighting reserve; a dedicated source (a fire truck or portable pump) carries no such limit. The model evaluates the rule as an event (a reserve violation) and as an adaptive flow limit.

3. Mathematical formulation

3.1 Hydraulic power and torque

The hydraulic power of a supply at pressure Δp and flow Q is

$$P_{\mathrm{h}} = \Delta p \, Q \tag{1}$$

and the torque at the valve stem of a motor of displacement D operating at a pressure drop Δp_{m} through a gearbox of ratio R is

$$T_{\mathrm{out}} = \frac{D \, \Delta p_{\mathrm{m}}}{2\pi} \, \eta_{\mathrm{m}} \, R \, \eta_{\mathrm{g}} \tag{2}$$

where η_{m} and η_{g} are the motor and gearbox efficiencies. The efficiency at breakaway and at stall is lower than the running efficiency because of static friction and start-up leakage, so the breakaway and stall torque uses $\eta_{\mathrm{m}} - \delta$, with δ a sampled loss; this is the mechanism by which the concept-stage model, which used one efficiency, overstated the stall torque. For the baseline D=200 cc, R=250, η_{m}=0.85, η_{g}=0.70 and Δp_{m}=100 psi, Eq. (2) gives 3,265 N.m.

3.2 Supply, hose and circuit pressure balance

The pressure available at the tap falls as flow is drawn. In the form used in hydrant flow tests [9], with a static pressure p_s and a flow capacity Q_a at the residual pressure p_r=20 psi,

$$p_{src}(Q) = p_s - (p_s - p_r)\left(\frac{Q}{Q_a}\right)^{1.85} \tag{3}$$

The hose loses pressure as $C\,(Q/100)^2\,(L/100)$ psi for a hose of length L in feet and a flow in gpm, with a coefficient C that depends on the hose size [8]. The coupling, the strainer and the exhaust hose add losses that are proportional to the square of the flow. The flow in operation is the largest flow at which the pressure balance

$$p_{src}(Q) - \Delta p_z - \left(\frac{CL}{10^6} + k_{spc} + \frac{c_f}{Q_{ref}^2} + \frac{C_d L_d}{10^6}\right)Q^2 \;\geq\; \Delta p_{run} \tag{4}$$

holds, limited by the flow limiter Q_{lim}, where Δp_z is the elevation head, k_{spc} the coupling coefficient, c_f the fouling loss at the reference flow Q_{ref}, $C_d L_d$ the exhaust-hose term, and $\Delta p_{run}=T_{run}/k_T$ the pressure drop that the running torque T_{run} requires. Eq. (4) is solved by bisection for every draw.

3.3 Motor speed, leakage, stroke time and water use

A water-hydraulic motor leaks through its clearances, and the leakage is nearly proportional to the pressure drop because the viscosity of water is low. The flow into the motor is therefore

$$Q = D\,n_m + c_L\,\Delta p_m, \qquad c_L = \left(\frac{1}{\eta_v} - 1\right)\frac{D\,n_{ref}}{\Delta p_{ref}} \tag{5}$$

where n_m is the motor speed, η_v the volumetric efficiency at the reference point (Δp_{ref}=100 psi, n_{ref}=100 rpm) and c_L the leakage coefficient. The output speed is $n_{out}=n_m/R$, the stroking time is $t=N_{turns}/n_{out}$ for N_{turns} handwheel turns, and the water used in one stroke is $V=Q\,t$. With creep flows Q_s and Q_e held for N_s and N_e turns, at output speeds n_s and n_e, the time is

$t=(N_{\text{turns}}-N_{\text{s}}-N_{\text{e}})/n_{\text{out}}+N_{\text{s}}/n_{\text{s}}+N_{\text{e}}/n_{\text{e}}$ and the water is the sum of the three phases. A stroke is required to be completed within 15 minutes of operating time, excluding set-up.

3.4 Relief valve, stall and the pressure surge

The relief valve opens at the set pressure p_{set}, which the chart derives from the valve class: $p_{\text{set}}=\mu\, T_{\text{ref}}/k_{\text{T}}^{\text{nom}}$ for a multiple μ of a reference torque T_{ref}, which is the median breakaway torque or the nominal allowable stem torque. The set point has a tolerance, and the relief valve overshoots by a fraction ε_{r} at full flow. When the motor stops abruptly, at the end of travel or against an obstruction, the moving column of water produces a surge, estimated by the Joukowsky relation for an instantaneous stop and reduced when the stop takes longer than the wave takes to travel the hose and return [16],

$$\Delta p_{\text{J}} = \rho\, a\, v \min\left\{1, \frac{2L}{a\, t_{\text{c}}}\right\}(1-\chi) \tag{6}$$

where a is the wave speed in the compliant hose, $v=Q_{\text{stall}}/A_{\text{h}}$ the velocity in the hose of cross-section A_{h} just before the stall, t_{c} the stopping time and χ the fraction absorbed by an accumulator. The flow before the stall is the end creep flow Q_{e} when the stroke proceeds to its end, the start creep flow Q_{s} when the valve does not break free (the tool then stalls at the start), and the running flow when there is no creep. The peak pressure at the motor is the smaller of the unrelieved surge and the relief override,

$$p_{\text{peak}} = \min\left\{p_{\text{s}} + \Delta p_{\text{J}},\ p_{\text{set}}(1+\varepsilon_{\text{r}}) + \kappa\, \Delta p_{\text{J}}\right\} \tag{7}$$

where κ is the fraction of the surge that the relief valve does not relieve in time. The peak torque is $T_{\text{peak}}=k_{\text{T}}^{\text{st}}\, p_{\text{peak}}$, with the stall constant k_{T}^{st} of Eq. (2) and the reduced efficiency. Without a relief valve $p_{\text{peak}}=p_{\text{s}}+\Delta p_{\text{J}}$. The surge model is a screening approximation: Section 4.3 checks its relations against a transient simulation, and an instrumented stall test is the validation that it still needs.

3.5 Events and the metrics reported

Five events are defined for every draw. A stroke succeeds if the tool delivers the breakaway torque ($T_{cap}=k_T^{st}\min\{\Delta p_s, p_{set}\}\geq T_{break}$), sustains the running torque, and completes the stroke within the time limit. An exceedance occurs when the peak stall torque T_{peak} is greater than the allowable stem torque T_{allow}, which is a design limit that already contains a safety factor. Damage occurs when T_{peak} is greater than T_{allow} times a sampled ratio between yield and allowable torque. A safe stroke is a successful stroke without exceedance. A reserve violation occurs on a shared source when the flow is greater than the spare flow. The concept-stage study reported a success probability of about 80% and, separately, an over-torque probability of about 5.5%; they were not combined, and the abstract of that study described the 80% as success without exceeding the allowable stem torque, which it was not. The safe stroke is the combined measure used here.

4. Probabilistic methodology

4.1 Inputs and sampling

Every input of the model is a random variable with an assumed distribution (Table 1). Three groups are sampled for each trial. The supply: the source type (firewater ring, fire truck, municipal hydrant) with its static pressure and flow capacity, the hose size and length, the elevation difference, the strainer fouling with a 3% chance of severe fouling, and the spare flow of shared sources. The valve: a lognormal breakaway torque by valve class, seizure events that multiply the torque by 1.5 to 3, a running torque of 30 to 60% of breakaway (the stem packing alone can add a force that is a sizeable part of the actuation load [17]), the number of handwheel turns to stroke, an allowable stem torque, and the ratio between yield and allowable torque. The tool: motor, volumetric and gearbox efficiencies, the loss of efficiency at stall, the tolerance and overshoot of the relief valve, and the wave speed, stopping time and unrelieved fraction of the

surge. The valve classes are a 12-inch gate valve, a 24-inch gate valve, a 36-inch butterfly valve, a 16-inch plug valve and a legacy seized valve, weighted 30, 25, 15, 10 and 20% in the reference mix; a separate class represents the 1,500 N.m valve of the industrial challenge [3].

Latin-hypercube sampling is used for the design search [18,19], and plain random sampling for every reported probability. All designs are evaluated on the same random draws (common random numbers [20]), so that differences between designs are not blurred by sampling noise. Correlations between inputs are not imposed, which is a limitation; rank-correlation methods are available if data on correlated inputs are obtained [21]. Probabilities are reported with Wilson 95% intervals, which describe only the Monte Carlo sampling error of the 200,000 draws of each case. They are not confidence intervals for the performance of a real tool, because the input distributions are assumptions.

Input	Distribution	Basis
Supply source	Firewater ring 50% (N 140, sd 20 psi); fire truck 30% (N 120, sd 25); municipal hydrant 20% (N 60, sd 15); clipped to 20–250 psi	Assumed
Source flow capacity at 20 psi residual	Lognormal, median 2,000 (ring), 500 (truck), 1,000 (hydrant) gpm; sigma 0.5, 0.4, 0.5	Assumed
Spare flow of a shared source	Lognormal, median 400 (ring), 200 (hydrant) gpm, sigma 0.6, 0.5; unlimited for a truck	Assumed
Hose	1.75 in 30%, 2.5 in 50%, 3 in 20%; length uniform 50–400 ft; friction coefficients 15.5, 2.0, 0.8	Fire-service practice [8]
Elevation, fouling	Elevation N(0, 15 ft); fouling lognormal median 2 psi, sigma 0.8, plus 3% chance of 20–60 psi	Assumed
Breakaway torque	Lognormal by class: gate 12 in 400 N.m (sigma 0.45); gate 24 in 1,200 (0.50); butterfly 36 in 1,500 (0.45); plug 16 in 800 (0.50); legacy 2,000 (0.60); challenge valve 1,500 (0.35)	Assumed; no published distribution was found
Seizure, running torque	Probability 5–25% by class, multiplier uniform 1.5–3; running torque 30–60% of breakaway	Assumed
Allowable stem torque	3 x class median, lognormal sigma 0.25; ratio of yield to allowable uniform 1.3–2.0	Assumed
Efficiencies	Motor N(0.85, 0.03); gearbox and interface N(0.70, 0.05); volumetric N(0.92, 0.02); loss at stall uniform 0.05–0.20	Typical ranges; assumed
Relief valve	Set-point tolerance sd 5%; override above the set point uniform 5–25%; unrelieved surge fraction uniform 0–30%	Assumed
Surge	Wave speed uniform 150–400 m/s; stopping time uniform 0.05–0.5 s	Assumed [16]

Table 1. Input distributions of the model (all assumed priors; none is a measurement).

4.2 Verification and its limits

The model is verified in five ways, none of which validates its physical assumptions. First, with all six added mechanisms switched off the engine reproduces the concept-stage model draw for draw. Second, with all random inputs fixed except the valve demand, the simulated breakaway probability agrees with the closed form for a mixture of lognormal valve classes to within 0.0024 over supply pressures of 40 to 200 psi (Fig. 2a), and the over-torque probability of a relief-limited tool agrees with its closed form to within 0.004. Third, the mechanical output power never exceeds the hydraulic power at the motor (maximum ratio 0.791). Fourth, 30 automated tests check unit conversions, the hose-friction form, the Joukowsky value, monotonic behavior in the supply pressure, the accumulator and the relief setting, and the creep logic. Fifth, convergence is checked by batch means: the standard error of the baseline probability of a safe stroke over ten blocks is 0.09 percentage points (Supplementary Fig. S1).

Numerical verification is distinguished from physical validation. The simulation is consistent with its own assumptions; whether those assumptions describe real valves, hoses, motors and relief valves can only be decided by measurement [22]. The input distributions in particular are not calibrated. For that reason the conclusions are tested against deliberately wrong distributions (Section 4.4).

4.3 Check of the surge model against a transient simulation

The surge relations of Eqs. (6) and (7) are classical, but the model applies them to an unusual system: a compliant hose between a flow-limited source and a stalling motor with a fast relief valve. They are therefore checked against a one-dimensional transient simulation of the hose by the method of characteristics [16], with a source behind a resistance (the flow limiter), a motor end that stops linearly in time, and a relief valve that opens linearly over a response time and

passes the running flow at its override pressure. It omits friction, hose-wall dynamics, cavitation, motor inertia and the true characteristic of a relief valve.

Three comparisons are made (Fig. 3). First, the solver reproduces the classical results: the Joukowsky surge for an instantaneous stop (26.0, 43.3 and 69.3 psi for wave speeds of 150, 250 and 400 m/s in a 2.5-inch hose at 60 gpm), the Michaud relation for slow stops, and the reduction factor of Eq. (6) for short stops. Second, for 300 random combinations of hose, flow, relief setting, stopping time and relief response time, the peak pressure of the screening model is at or above the simulated peak in 97% of the cases, with a median ratio of 1.19. Third, the unrelieved fraction κ implied by the simulation is below the upper bound of the prior (0.3) for relief valves that respond within 0.1 s (95th percentile 0.03), but not for slower ones: at 0.3 s the 95th percentile is 0.45 and 11% of the cases exceed the bound. The response time of the relief valve is therefore a requirement of the design: the results of Section 6.6 hold only for a relief valve that responds within about 0.1 s.

The check compares the model's relations with a more detailed model of the same physics; it is not an experimental validation. No experimental record of a stall surge in a fire hose behind a water-hydraulic motor was found in a search that was not exhaustive, so the surge model is a screening approximation that is verified, not validated, and the instrumented stall test of Section 7.4 is the experiment that it needs.

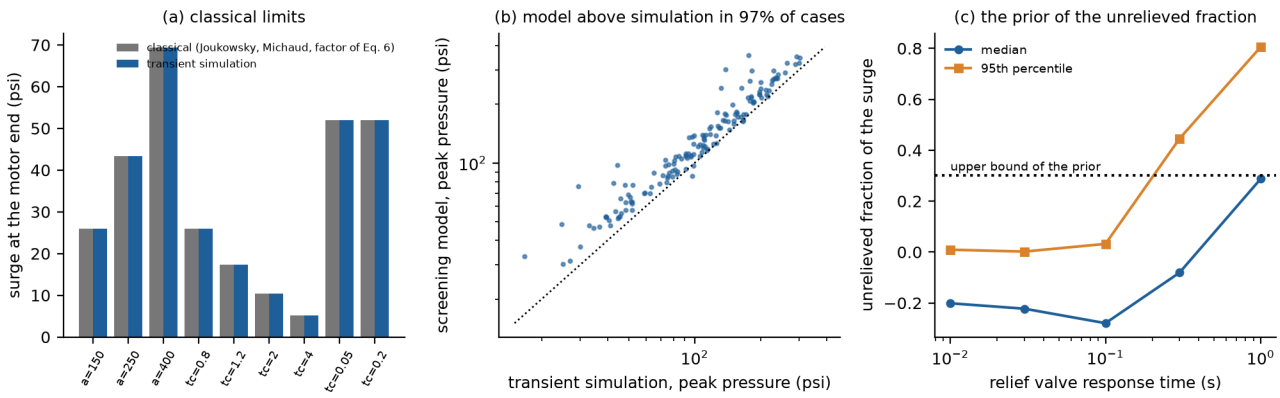


Fig. 3. Check of the surge model. (a) Surge at the motor end: classical relations (Joukowsky for an instantaneous stop at wave speeds of 150, 250 and 400 m/s; Michaud for slow stops; the reduction factor of Eq. (6) for short stops) and the transient simulation. (b) Peak pressure of the screening model against the peak pressure of the simulation for random hoses, flows, relief settings and valve response times; the dotted line is equality. (c) Unrelieved fraction of the surge implied by the simulation against the response time of the relief valve; the dotted line is the upper bound of the prior of the model.

4.4 Assumption shifts

Eight shifts are applied to the draws, one at a time, and then a combined stress, to see whether a conclusion survives a wrong prior: breakaway torques multiplied by 1.5; the scatter of the breakaway torque increased by 0.15 in log units; supply pressure lowered by 30 psi; seizure probabilities doubled; motor and gearbox efficiencies lowered by 0.10; fouling tripled; allowable stem torque multiplied by 0.7; the stall surge doubled; and all of them at a reduced strength together. The shifts change the distributions, not the design, so the same design can be evaluated under each; the worst case over the shifts is a robustness measure [23].

5. Multi-objective optimization

The design problem has four variables: the displacement D (60 to 500 cc), the ratio R (60 to 600), the relief multiple μ (0.2 to 1.0 of the nominal allowable stem torque, or 0.6 to 4 times the class median breakaway torque) and the flow limit Q_{lim} (15 to 100 gpm), plus, when creep is allowed, the creep flows at the start and end of travel (5 to 60 gpm, held for one and two turns). The adaptive reserve rule is applied, and a design is admitted only if the motor speed at its flow limit is at most 3,000 rpm, an assumed requirement (no published speed limit of water-hydraulic motors of this class was found); the best designs without that limit, and with a limit of 1,500 rpm, are reported for comparison. Eight families are searched: two relief bases, each without and with an accumulator that absorbs 75% of the surge, each without and with creep. The objectives are to maximize the probability of a safe stroke and to minimize the probability of exceedance and the median water used in a stroke, under the decision constraint

$$P(T_{\text{peak}} > T_{\text{allow}}) \leq 0.01 \tag{8}$$

The limit of 1% is an illustrative research constraint, not an established acceptable industrial safety limit; Section 6.6 shows how the result moves with it. Each family is searched with 1,500 Latin-hypercube designs on 15,000 common draws; the non-dominated set is found by exhaustive comparison [24,25]. The search is a sampling method, not a proof of optimality. To control the optimizer's curse the chosen designs are re-evaluated on 200,000 independent draws, and a robust design is chosen by maximizing the worst-case probability of a safe stroke over the assumption shifts subject to a 2% limit on the worst-case over-torque probability.

A design chosen because its estimated over-torque probability is at most 1% has an uncertain estimate, and the selection favors designs whose estimate happened to fall below the limit.

Wilson 95% limits are therefore reported on the independent draws (Table 7), together with a strict rule that admits a design only if the upper limit of its in-sample estimate is at most 1%.

6. Results

6.1 The baseline tool and what the added mechanisms change

The baseline tool has a 200 cc motor, a 250:1 gearbox, a 60 gpm flow limiter and a relief valve set at twice the median breakaway torque of the valve class, and is evaluated on 200,000 draws. Under the assumed priors it completes the stroke in 75.3% of the draws (Wilson interval 75.1 to 75.5), but it does so without exceeding the allowable stem torque in only 62.6% (62.4 to 62.8). The stall torque exceeds the allowable value in 15.2% of the draws and the yield margin in 1.1% (Table 2). The concept-stage model gave a success probability of 80.0%, an over-torque probability of 5.3% and, when the two are combined, 75.3%. Model 2 gives 62.6%: about five eighths of the draws, not four fifths, give a safe stroke, and the surge on stall, which the concept-stage model omitted, is the main reason the over-torque probability now exceeds 15%.

Valve class	Success v1	Success	Safe stroke	Over-torque	Stem damage	Time (min)	Water (L)
Gate 12 in	90.3%	84.2%	60.9%	27.0%	2.8%	4.0	900
Gate 24 in	84.0%	77.1%	68.0%	11.1%	0.3%	8.8	2,002
Butterfly 36 in	66.4%	72.5%	65.1%	9.3%	0.3%	11.4	2,578
Plug 16 in	86.8%	80.2%	67.6%	15.1%	0.7%	2.6	600
Legacy seized	66.2%	58.7%	53.8%	6.8%	0.2%	9.4	2,128
Weighted mix	80.0%	75.3%	62.6%	15.2%	1.1%	7.1	1,601

Table 2. Baseline tool (200 cc, 250:1, 60 gpm, relief at 2 x class median breakaway). Success of the concept-stage model (v1) and of model 2; probabilities over 200,000 draws per class; time and water are medians of successful strokes. Safe stroke = success without exceedance of the allowable stem torque.

The six added mechanisms do not change the result equally (Table 3, Fig. 2b). The source characteristic, the flow-dependent fouling loss and the exhaust back-pressure change the success

probability by less than 0.2 percentage points, because the flow limiter, not the circuit, sets the flow in most draws (the median operating flow is 60 gpm, the limiter setting). The lower torque efficiency at breakaway and stall lowers the success probability by 6.7 points and the over-torque probability by 3.8 points, since the stall torque is smaller than the concept-stage model assumed. Replacing the constant volumetric factor by a leakage that is proportional to the pressure drop raises the success probability by 1.7 points (the motor leaks less at the low running pressures than the constant factor implied). The surge on stall changes the picture: it raises the over-torque probability from 1.6% to 15.4% and the damage probability from 0.02% to 1.1%.

Step	Success	Safe stroke	Over-torque	Stem damage	Interference (v1 metric)
v1 physics	80.0%	75.3%	5.5%	0.09%	31.3%
+ flow supply	80.0%	75.3%	5.5%	0.09%	31.3%
+ foul flow	80.2%	75.5%	5.5%	0.09%	31.4%
+ back	80.2%	75.5%	5.5%	0.09%	31.4%
+ start eff	73.6%	72.2%	1.6%	0.02%	31.4%
+ leak	75.3%	73.9%	1.6%	0.02%	31.4%
+ surge	75.3%	62.3%	15.4%	1.08%	31.4%

Table 3. Switching the model-2 mechanisms on one at a time, in order, on the same 60,000 draws (baseline tool, valve-class mix).

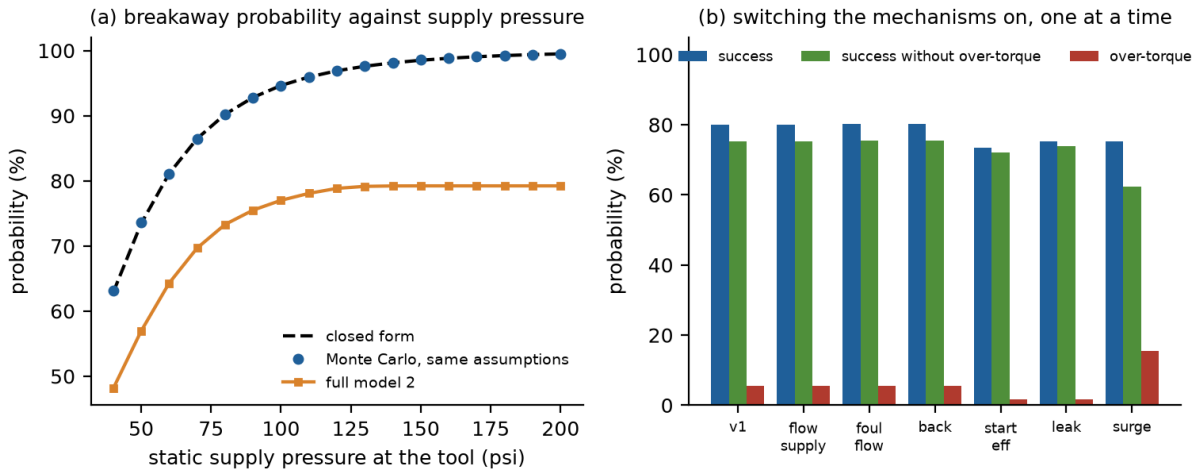


Fig. 2. Verification and mechanism ladder. (a) Breakaway probability against the supply pressure at the tool: closed form, Monte Carlo with the same assumptions, and the full model.

(b) Success, success without over-torque and over-torque probability as the mechanisms of model 2 are switched on in order (v1: the concept-stage physics).

The concept-stage study stated that the tool draws 50 to 100 L/min (13 to 26 gpm), and its simulation used a 60 gpm (227 L/min) flow limiter. The 13 to 26 gpm is the flow that a 200 cc motor needs at an output speed of 1 to 2 rpm. Valves that need 30 to 70 handwheel turns cannot be stroked in 15 minutes at that speed: the median output speed of a successful stroke in the simulation is 4.5 rpm. The operating flow is therefore the limiter setting, 60 gpm (227 L/min), in most draws, and a successful stroke uses a median of 1,601 L of water (90th percentile 2,632 L; Fig. 14b). A limit of 20 gpm completes the stroke within 15 minutes in only 40% of the draws, and 40 gpm in 73% (Fig. 8a).

6.2 The torque limiter, the surge on stall and stem damage

The relief valve is essential. Without it the success probability is 86.5%, but the success without exceedance of the allowable stem torque is only 19.1% and the probability of damage (a torque above the yield margin) is 44.9%. With the relief valve set at twice the median breakaway torque the safe-stroke probability is 62.7% (Fig. 5a). Lowering the setting to 1.5 times the median reduces the over-torque probability to 5.1% while the safe-stroke probability falls to 57.5%; at a setting equal to the median breakaway torque the safe-stroke probability is 32.9% because the tool can no longer break half of the valves free. The success probability and the over-torque probability both rise with the setting, so the probability of a safe stroke has an interior maximum near twice the median; the setting is a safety-critical, valve-specific parameter, and whatever safety the chart buys is paid for in valves that the tool cannot move.

The surge on stall is the largest single source of over-torque in the model. Its median is 45 psi (40% of the static supply pressure), with a 95th percentile of 119 psi. For a small valve the relief

pressure is low: a 12-inch gate valve with a median breakaway torque of 400 N.m has a relief set-point of about 25 psi on the baseline tool, below the typical surge, so the pressure that the relief valve cannot relieve in time exceeds the pressure that the chart is meant to enforce; the over-torque probability of this class is 27% (Table 2). A hose-side accumulator that absorbs a quarter, a half or three quarters of the surge lowers the over-torque probability of the baseline tool from 15.2% to 12.3%, 9.5% and 6.9%, and the damage probability from 1.11% to 0.18% (Fig. 5b). The surge inputs, the wave speed in the hose, the stopping time and the unrelieved fraction, are assumed, and Fig. 12a shows how the over-torque probability follows the surge multiplier: when the surge is twice the prior the over-torque probability is 26% for the baseline tool, 10% with the accumulator, and 2.3% for the optimized design of Section 6.6.

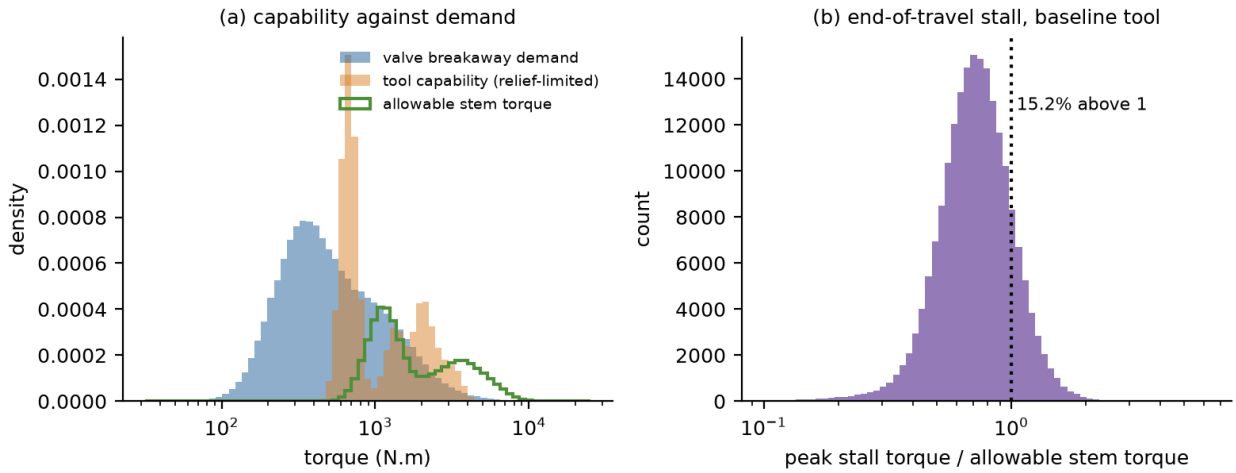


Fig. 4. (a) Distributions of the valve breakaway demand, the stall capability of the baseline tool (limited by the relief valve) and the allowable stem torque over the draws. (b) Ratio of the peak stall torque to the allowable stem torque; the dotted line marks the exceedance.

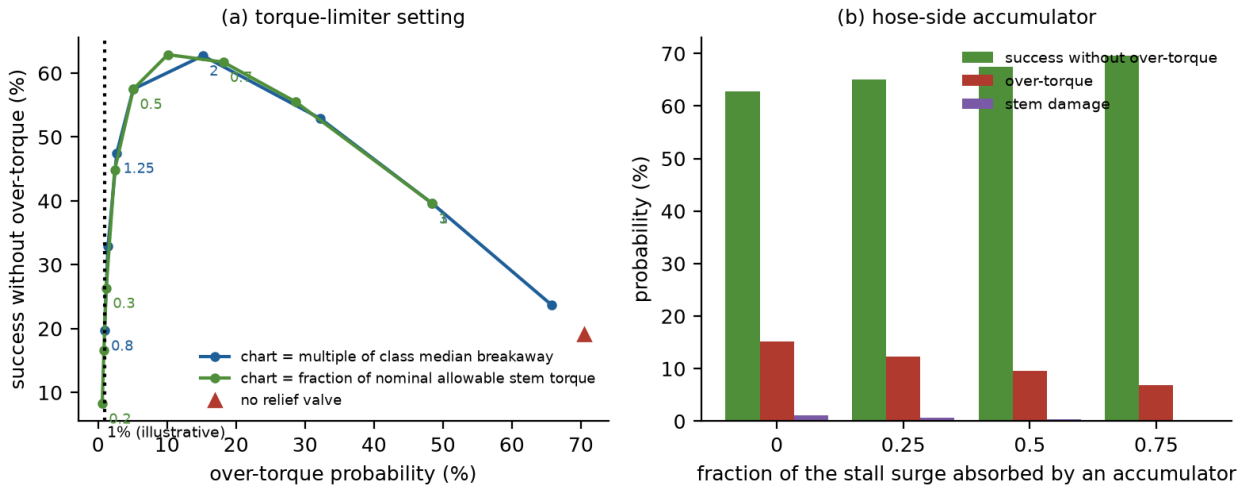


Fig. 5. (a) Probability of a safe stroke against the probability of over-torque as the relief setting is varied (labels: multiple of the median breakaway torque, or fraction of the nominal allowable stem torque), for the two chart bases and without a relief valve. (b) Effect of a hose-side accumulator on the safe stroke, the over-torque and the stem damage.

6.3 Creep at the start and at the end of travel

A tool whose flow can be turned down is a natural remedy for the surge, which is proportional to the velocity of the water that the stall has to stop. Two creep phases are evaluated for the baseline tool: a slow flow for the first turns of the stroke, which builds torque gradually against a stuck valve, and a slow flow for the last turns, which brings the tool to the stop slowly. In the model the end creep lowers the flow before the stall, and the start creep lowers it for valves that stall at the start because they do not break free; both cost time, and a stroke that no longer finishes in 15 minutes is a failure. The opening transient of the supply is not modeled; the slow-opening shut-off is assumed to remove it.

Accum.	Start	End	Turns	Safe	Over-torque	Damage	Success	Time	Water
none	-	-	-	62.7%	15.2%	1.11%	75.5%	7.1	1,609
none	-	40	1.0 + 2.0	65.6%	12.0%	0.61%	75.2%	7.3	1,603
none	-	20	1.0 + 2.0	67.6%	8.9%	0.34%	74.1%	7.9	1,583
none	-	10	1.0 + 2.0	66.4%	7.6%	0.28%	71.4%	9.0	1,528

Accum.	Start	End	Turns	Safe	Over-torque	Damage	Success	Time	Water
none	20	-	1.0 + 2.0	62.1%	14.0%	0.98%	74.8%	7.5	1,597
none	10	-	1.0 + 2.0	61.2%	13.8%	0.97%	73.8%	8.1	1,576
none	20	20	1.0 + 2.0	67.0%	7.8%	0.21%	73.4%	8.2	1,570
none	10	10	1.0 + 2.0	63.1%	6.2%	0.15%	68.0%	9.3	1,343
none	10	10	0.5 + 1.0	67.8%	6.2%	0.15%	72.9%	8.5	1,560
none	10	10	2.0 + 4.0	42.3%	6.2%	0.15%	45.9%	10.7	935
none	10	10	4.0 + 8.0	28.6%	6.2%	0.15%	31.3%	12.5	819
0.75	-	-	-	69.6%	6.9%	0.18%	75.5%	7.1	1,609
0.75	-	20	1.0 + 2.0	69.5%	5.7%	0.12%	74.1%	7.9	1,583
0.75	10	10	1.0 + 2.0	63.9%	5.2%	0.11%	68.0%	9.3	1,343

Table 4. Creep at the start and at the end of travel, baseline tool (200 cc, 250:1, 60 gpm, relief at 2 x median), 60,000 draws: flows (gpm), turns at those flows, safe stroke, over-torque, damage, success, median time of successful strokes (min) and water (L). First column: accumulator fraction.

Creep at the end of travel works. Two turns at 20 gpm lower the over-torque probability from 15.2% to 8.9% and the damage probability from 1.11% to 0.34%, and raise the probability of a safe stroke from 62.7% to 67.6%, at the cost of 0.8 minutes of median stroking time and 1.4 points of success (strokes that no longer finish in 15 minutes). A slower approach, 10 gpm, lowers the over-torque probability to 7.6% but the safe-stroke probability to 66.4% (success 71.4%). Creep at the start alone does little for the over-torque probability (14.0% at 20 gpm and 13.8% at 10 gpm, against 15.2%) because in this model it matters only for valves that do not break free; its practical benefit, a shock-free build-up of torque against a stuck valve, is not captured. The length of the slow phases matters as much as their flow (Fig. 6c): two and four turns at 10 gpm raise the median time to 10.7 minutes and cut the success to 45.9%. With the accumulator the gain of the end creep is smaller (over-torque from 6.9% to 5.7%), so the two measures are complementary with diminishing returns.

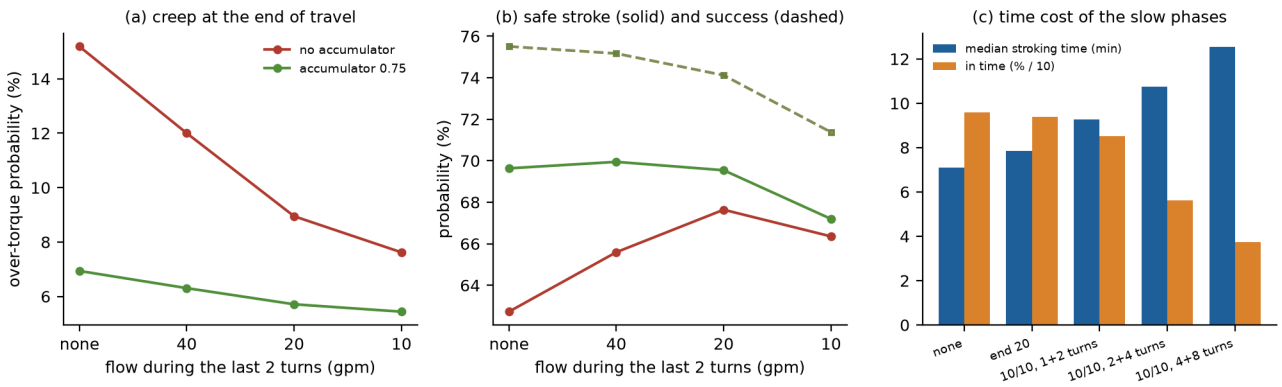


Fig. 6. Creep at the start and at the end of travel, baseline tool. (a) Over-torque probability against the flow during the last two turns, without and with an accumulator. (b) Probability of a safe stroke (solid) and of success (dashed). (c) Median stroking time and share of strokes finishing in 15 minutes (divided by 10) for slow phases of different flow and length.

6.4 The design space

With the flow limited, the torque scales with $D\,R$ and the speed falls in the same proportion, so useful designs lie along a ridge of constant $D\,R$ (Fig. 7a): 62% or more of the draws give a safe stroke along the diagonal from 100 cc at 500:1 to 400 cc at 100:1, and 7% for the largest products. The over-torque probability rises with $D\,R$ (Fig. 7b, from 2% to 42% over the grid) because a larger product means a lower relief pressure, and the surge, which is a pressure, becomes a larger fraction of it.

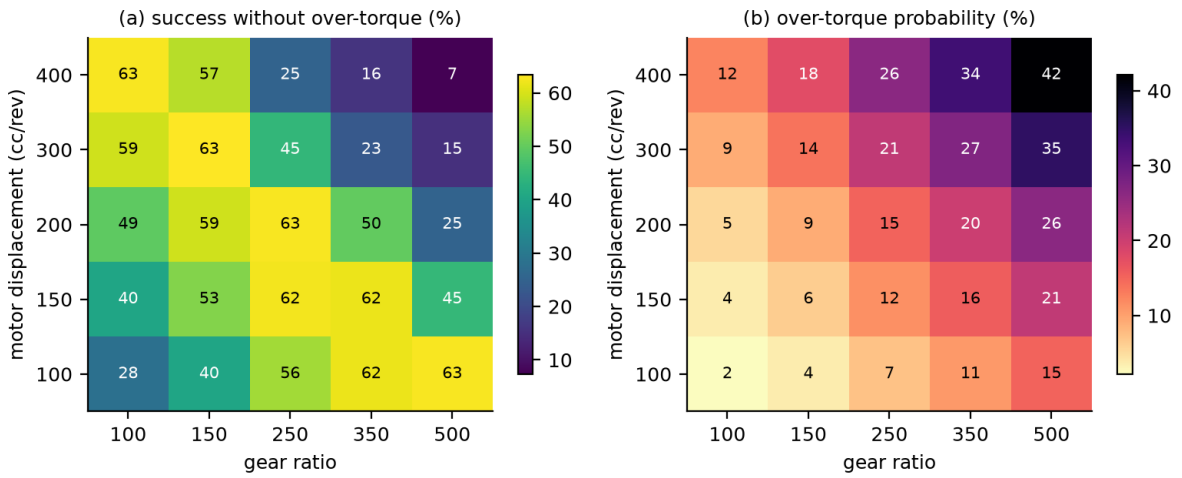


Fig. 7. (a) Probability of a safe stroke and (b) probability of over-torque against motor displacement and gear ratio, for the baseline relief chart (2 x median) and 60 gpm.

6.5 Firewater reserve and the flow limit

The concept-stage study found that the tool draws more than a quarter of the available source flow in 31% of the draws. That figure depends on how the source flow is defined and is 32% in the present model, where it is kept for continuity (Table 3). The event that matters for safety is different: the tool takes more than the flow a shared source can spare without cutting its firefighting reserve. Under the assumed spare flows (median 400 gpm for a firewater ring and 200 gpm for a hydrant) this reserve violation occurs in only 0.2% of the draws at 60 gpm and in 2.7% at 150 gpm, but it is very sensitive to the assumption: if the spare flow is a quarter of the prior it occurs in 21%, and if it is a tenth in 55% (Fig. 8c). The adaptive rule, which lowers the limiter to the spare flow of a shared source, removes the violations by construction and costs success: the safe-stroke probability falls from 62.3% to 60.0% with a quarter of the spare flow, and to 48.2% with a tenth, where only 71% of the strokes finish in 15 minutes. A site that has not measured its spare firewater flow cannot be assured that the tool is compatible with its firefighting reserve, and the right operating rule is to use a dedicated source for large valves.

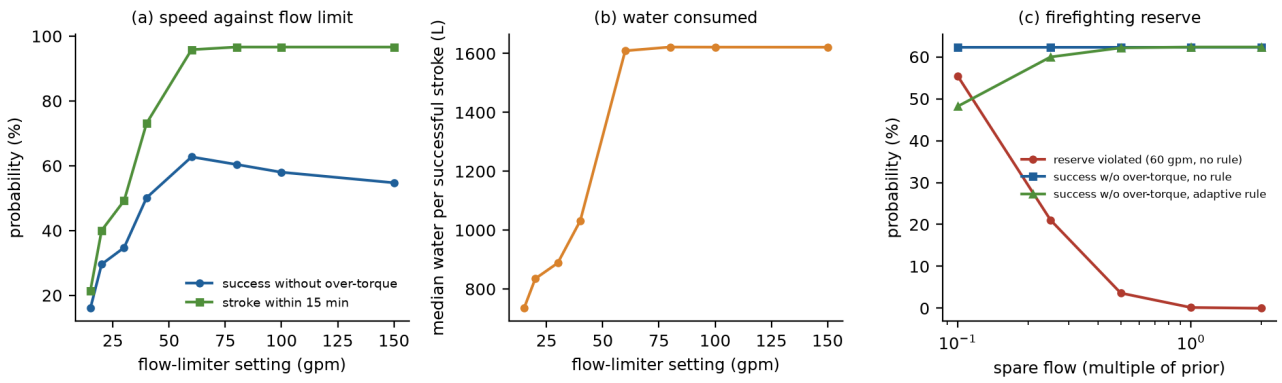


Fig. 8. (a) Probability of a safe stroke and of finishing within 15 minutes against the flow limit. (b) Median water used in a successful stroke against the flow limit. (c) Probability of violating the firefighting reserve at 60 gpm, and the probability of a safe stroke without and with the adaptive rule, against the spare flow of shared sources (multiple of the prior).

6.6 Safety-constrained optimization

Design	D	R	Relief	Q	rpm	Creep	Safe	Over-torque	Damage	Water (L)
Baseline tool	200	250	2.0 M	60	1,136	none	62.6%	15.2%	1.06%	1,601
Chart from allowable torque, no accumulator	305	100	0.50 A	33	414	none	53.8%	1.0%	0.01%	906
Chart from median breakaway, no accumulator	203	148	1.51 M	29	548	none	51.4%	0.9%	0.01%	656
Chart from allowable torque, accumulator 0.75	135	319	0.50 A	85	2,378	none	59.6%	0.9%	0.01%	1,395
Chart from median breakaway, accumulator 0.75	141	217	1.58 M	75	2,006	none	58.4%	0.9%	0.01%	944
Chart from allowable torque, creep, no accumulator	270	160	0.51 A	70	978	30 / 5	59.5%	0.9%	0.01%	1,366
Chart from median breakaway, creep, no accumulator	392	100	1.56 M	59	571	16 / 7	60.6%	0.8%	0.00%	1,240
Chart from allowable torque, creep, accumulator 0.75	473	109	0.52 A	67	534	52 / 32	63.4%	1.0%	0.01%	1,689
Chart from median breakaway, creep, accumulator 0.75	447	115	1.58 M	73	616	33 / 16	64.0%	0.9%	0.01%	1,676

Table 5. Best design of each family under P(over-torque) <= 1%, found on 15,000 draws and re-evaluated on 200,000 independent draws (shown). Relief: multiple of the allowable stem torque (A) or of the median breakaway torque (M); D in cc, flow limit and creep flows (start / end) in

gpm, rpm the motor speed at the flow limit (limited to 3,000); water is the median per successful stroke (L).

A design that satisfies the 1% constraint exists in every family (316 to 690 of the 1,500 designs searched meet it), and the constraint is respected out of sample (Table 5): the over-torque probability of the chosen designs, re-evaluated on independent draws, is between 0.9% and 1.0%. The price is the probability of a safe stroke. Without an accumulator the best design completes a safe stroke in 53.8% of the draws, against 62.6% for the baseline tool, which exceeds the allowable stem torque in 15%; with an accumulator that absorbs three quarters of the surge the best design reaches 59.6%. The baseline tool with the accumulator and the original chart gives 69.5% with 6.9% over-torque, so at the 1% constraint the safer configuration gives up 3.1 to 8.8 percentage points of the safe-stroke probability of the baseline against a reduction of the over-torque probability by a factor of about 17. The chosen designs have a relief setting at about half of the nominal allowable stem torque (0.50 to 0.50 of it, which is 1.5 to 1.5 times the median breakaway torque), flow limits of 29 to 85 gpm and a product $D\,R$ of 30,026 to 43,172: lower than the baseline's 50,000. The probability of a safe stroke at this constraint is not sensitive to the exact displacement and ratio along the ridge (Fig. 9a).

Allowing the creep flows at the start and at the end of travel to be chosen raises the best probability of a safe stroke under the same constraint (Table 5, out of sample) from 53.8% to 59.5% without an accumulator and from 59.6% to 63.4% with one. The gain comes from the end of travel: the chosen designs run at 5 and 32 gpm for the last two turns, which is 8% and 49% of their flow limit, so the flow that the stall has to stop, and with it the surge, is reduced, and the flow limit and the relief chart can be set higher for the same over-torque probability (relief at 0.51 and 0.52 of the allowable stem torque, flow limits of 70 and 67 gpm). Without an

accumulator the end creep is 5 gpm and costs time (median 9.5 minutes against 7.0 without creep); with an accumulator it is 32 gpm. The start creep chosen, 30 and 52 gpm, should not be read as an optimum: in this model it acts only on valves that do not break free, so the objective is nearly flat in it.

The 1% limit is illustrative, and Table 6 shows how the best safe-stroke probability found by the search moves with it: for the family with the accumulator it is 56.5% at 0.5%, 60.0% at 1%, 65.5% at 2% and 68.5% at 5%. The decision belongs to the operator and the regulator, and the figures give them the exchange rate.

Family	P(over-torque) <= 0.5%	<= 1%	<= 2%	<= 5%
Chart from allowable torque, no accumulator	43.3%	54.0%	57.5%	62.7%
Chart from allowable torque, accumulator 0.75	56.5%	60.0%	65.5%	68.5%
Chart from allowable torque, creep, no accumulator	53.4%	59.5%	63.5%	68.6%
Chart from allowable torque, creep, accumulator 0.75	58.9%	63.9%	66.7%	70.2%

Table 6. Best probability of a safe stroke among the searched designs (in-sample) as the limit on the over-torque probability is varied.

Family	Chosen at estimate <= 1%	Safe	Strict rule	Safe	Qualifying
Chart from allowable torque, no accumulator	0.99% (0.95 to 1.03)	53.8%	0.55% (0.52 to 0.58)	49.6%	327
Chart from median breakaway, no accumulator	0.88% (0.84 to 0.92)	51.4%	0.75% (0.71 to 0.79)	47.6%	234
Chart from allowable torque, accumulator 0.75	0.92% (0.88 to 0.96)	59.6%	0.60% (0.57 to 0.64)	58.4%	565
Chart from median breakaway, accumulator 0.75	0.93% (0.89 to 0.98)	58.4%	0.75% (0.72 to 0.79)	57.6%	404
Chart from allowable torque, creep, no accumulator	0.87% (0.83 to 0.91)	59.5%	0.80% (0.76 to 0.84)	57.5%	415
Chart from median breakaway, creep, no accumulator	0.79% (0.75 to 0.83)	60.6%	0.79% (0.75 to 0.83)	60.6%	314
Chart from allowable torque, creep, accumulator 0.75	0.98% (0.93 to 1.02)	63.4%	0.70% (0.67 to 0.74)	62.3%	592
Chart from median breakaway, creep, accumulator 0.75	0.91% (0.87 to 0.95)	64.0%	0.73% (0.69 to 0.77)	61.7%	445

Table 7. Over-torque probability with Wilson 95% limits on 200,000 independent draws, for the best design of each family chosen at an estimated probability of at most 1% and for the best design chosen so that the upper limit of the in-sample estimate (15,000 draws) is at most 1%; probability of a safe stroke; number of designs that qualify under the strict rule.

On 200,000 independent draws the designs chosen at an estimated 1% have over-torque probabilities of 0.79% to 0.99%, and the upper 95% limit exceeds 1% for 2 of the eight (Table 7): a point estimate near 1% does not establish compliance with a strict limit. Under the strict rule every design has an upper limit of at most 0.84%, and the price is 0.0 to 4.2 percentage points of the probability of a safe stroke. A statement of compliance should be made with the strict rule, and with a number of draws chosen for the margin that is wanted; the limits are Monte Carlo limits for the assumed priors and say nothing about the error of the priors.

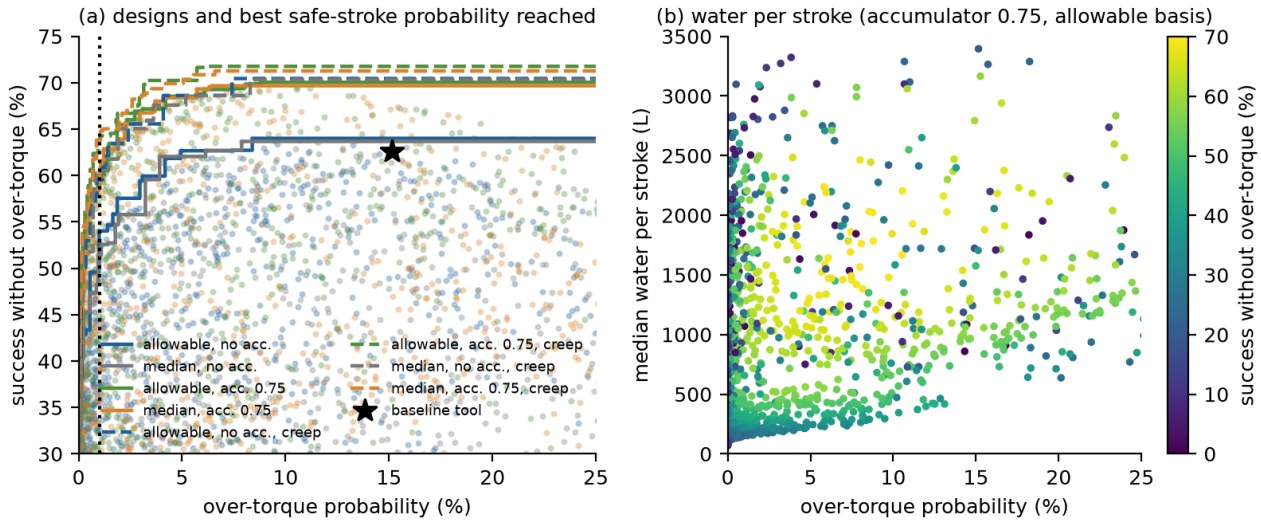


Fig. 9. (a) Searched designs (points) and non-dominated sets (lines) in the plane of safe-stroke probability against over-torque probability, for the eight families; dashed lines are the families with creep, the star is the baseline tool and the dotted line the 1% constraint. (b) Median water per stroke of the designs of the family with the accumulator and the allowable-torque chart.

A single configuration is almost as good as valve-class-specific ones: choosing the displacement, ratio, relief setting and flow limit separately for each class under the same constraint gives a weighted safe-stroke probability of 62.2% against 59.6% for the best single design with the accumulator (a gain of 2.6 points, part of it the optimism of choosing the best of 500 designs per class). What cannot be shared is the relief setting, which is read from the chart of each class (Supplementary Fig. S2).

6.7 Robustness to wrong input distributions

Assumption shift	Baseline	Baseline + accumulator	Optimized, no accumulator	Optimized, accumulator	Optimized, accumulator and creep
nominal priors	63% / 15.0%	69% / 6.8%	54% / 1.0%	59% / 0.9%	63% / 1.0%
valve torque x1.5	41% / 14.8%	46% / 6.8%	28% / 1.0%	31% / 0.9%	35% / 1.0%
valve scatter +0.15	59% / 14.9%	65% / 6.8%	52% / 1.0%	57% / 0.9%	60% / 1.0%
supply -30 psi	57% / 13.8%	63% / 5.7%	46% / 0.9%	54% / 0.8%	58% / 0.8%
seizure x2	58% / 14.9%	65% / 6.8%	49% / 1.0%	55% / 0.9%	58% / 1.0%
efficiencies -0.10	55% / 2.9%	57% / 0.6%	34% / 0.0%	39% / 0.0%	42% / 0.0%
fouling x3	63% / 14.9%	69% / 6.8%	54% / 1.0%	59% / 0.9%	63% / 1.0%
stem strength x0.7	33% / 53.0%	44% / 38.7%	46% / 12.2%	51% / 12.5%	54% / 13.8%
surge x2	54% / 25.6%	67% / 9.3%	53% / 2.6%	59% / 2.2%	63% / 1.5%
combined stress	33% / 20.3%	40% / 8.2%	25% / 1.8%	29% / 1.6%	33% / 1.5%

Table 8. Probability of a safe stroke / probability of over-torque under each assumption shift, for the baseline tool, the baseline with an accumulator, and the best designs of the allowable-torque families, with and without creep (120,000 draws each).

The optimum is not robust to every wrong prior (Table 8, Fig. 10). At the assumed priors the baseline tools have the larger probability of a safe stroke, but they pay for it in over-torque; the optimized designs keep the over-torque probability near 1% under shifts of the valve torque, the supply, the seizure, the efficiencies and the fouling, because the chart limits the torque whatever the demand, and the safe-stroke probability moves as the demand moves, for example from 59% to 31% when the breakaway torques are multiplied by 1.5. One shift breaks the constraint decisively. If the true stem strength is 30% lower than the data sheet says, the over-torque

probability of the optimized design with the accumulator rises to 13% (14% with creep as well), because the chart was computed from the nominal strength; a surge twice the prior roughly doubles it, to 2.2%, against 26% for the baseline tool. With the stems 30% weaker the order of the designs by safe strokes reverses: the optimized designs then lead, because the baseline tools lose most of their safe strokes to over-torque. These are the two inputs of which the model knows least, and they are the first things to measure. A chart that is safe against both would have to be lower, at a cost in the safe-stroke probability: among 150 candidate designs with a nominal over-torque probability of at most 2%, the largest worst-case safe-stroke probability that keeps the worst-case over-torque probability at or below 3%, 5% and 10% is 21%, 23% and 29% for the family with the accumulator (16%, 19% and 23% without it).

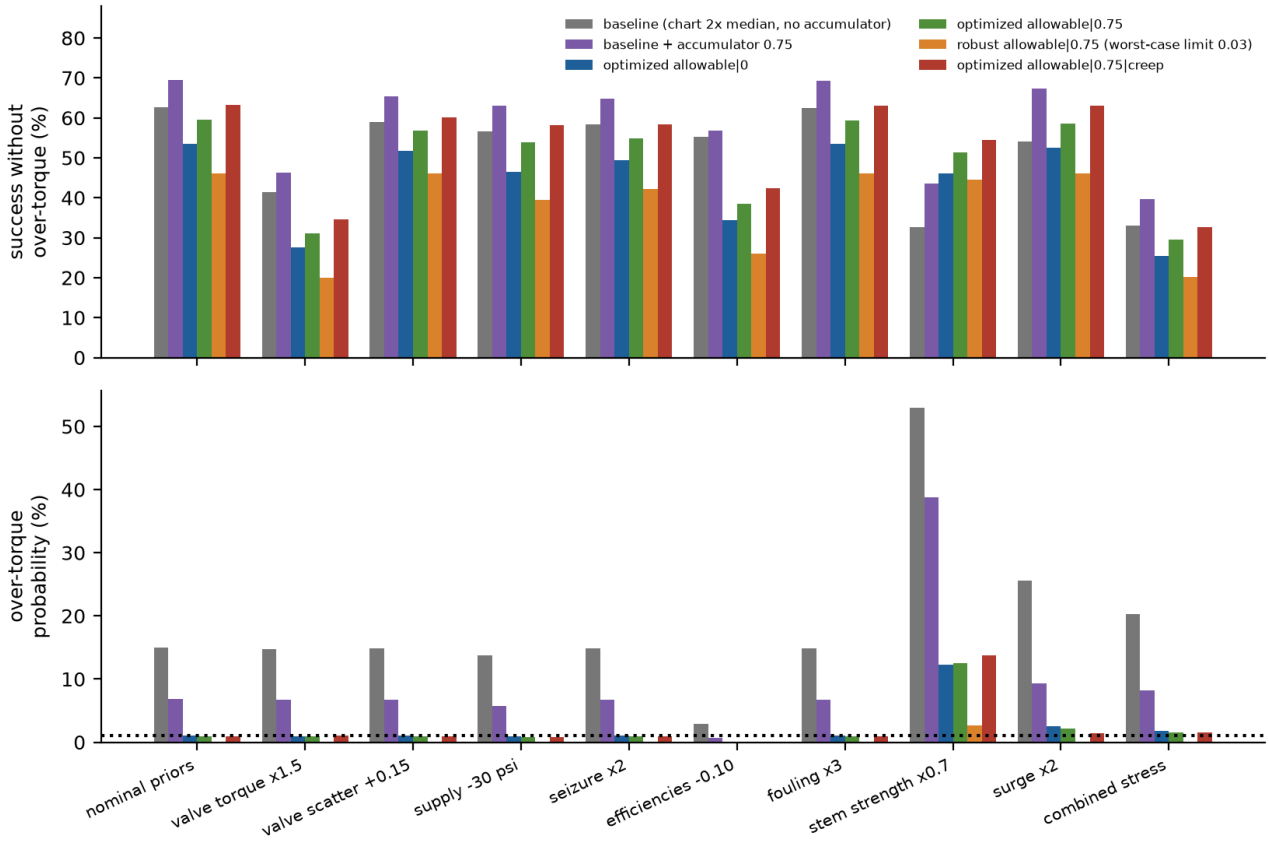


Fig. 10. Probability of a safe stroke (top) and of over-torque (bottom) under the assumption shifts for the baseline tool, the baseline with an accumulator, and the best designs found with and without an accumulator. The dotted line marks 1%.

The strongest limitation of the nominal 1% constraint is the stem strength, and it deserves a direct statement: the optimized configuration cannot be considered operationally safe without verified stem-strength information. The over-torque probability of the optimized design with an accumulator and creep is 0.9% when the stems are as strong as the data sheet says, 2.4% when they are 10% weaker, 6.0% when they are 20% weaker and 14% when they are 30% weaker (Fig. 11a); the baseline tool is at 15%, 24%, 37% and 53%. The chart is computed from the nominal strength, so an error in the nominal strength passes straight to the stem. Holding 1% over-torque when the stems may be weaker than the data sheet by a given factor requires a lower

chart (Table 9): 0.47 of the allowable stem torque instead of 0.53 when 10% weaker stems must be tolerated, 0.41 at 20% and 0.36 at 30%, and the probability of a safe stroke that remains falls to 57%, 48% and 38% (Fig. 11b). Each 10% of stem strength that is not known therefore costs about 7 to 10 percentage points of safe strokes. The operating rule that follows is that a relief chart is issued for a valve only with a verified stem strength for that valve, and that a valve of unknown stem strength is operated with a derated chart, or by another means.

Stem strength f	Chart	Factor	Safe stroke	Over-torque
1.0	0.53	1.02	63.8%	1.00%
0.9	0.47	0.91	56.9%	1.00%
0.8	0.41	0.80	48.2%	1.00%
0.7	0.36	0.69	37.8%	1.00%
0.6	0.30	0.57	26.0%	1.00%
0.5	0.24	0.46	14.8%	1.00%

Table 9. Relief chart (multiple of the nominal allowable stem torque) that keeps the over-torque probability at 1% when the true stem strength is a factor f of the data-sheet value, for the optimized design with an accumulator and creep (D = 473 cc, R = 109, flow limit 67 gpm; nominal chart 0.52), with the factor by which the nominal chart is lowered and the probability of a safe stroke that remains.

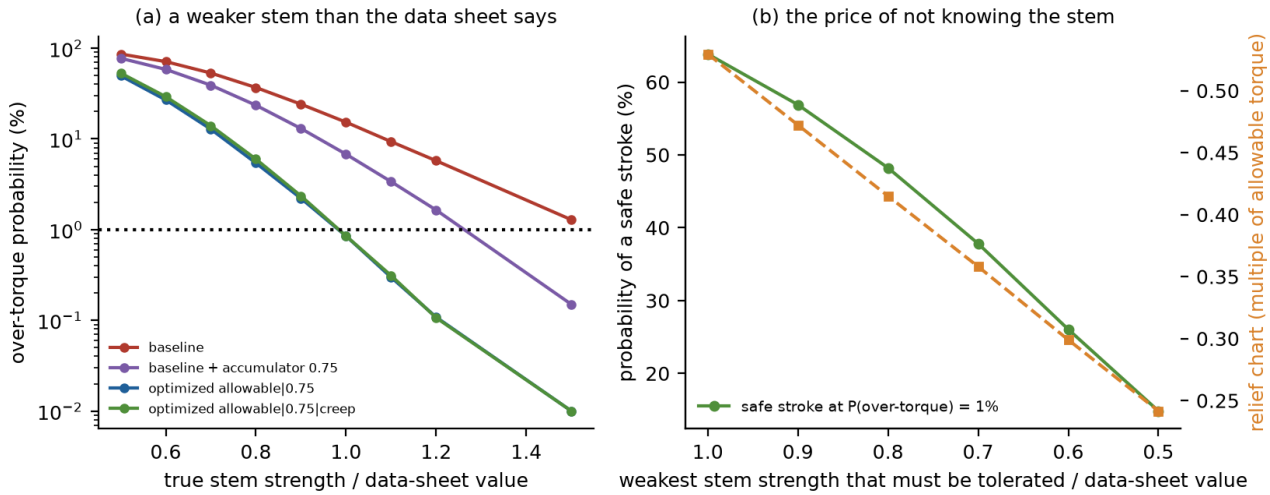


Fig. 11. (a) Over-torque probability against the true stem strength relative to the data-sheet value (the chart is computed from the data-sheet value), for the baseline tool, the baseline with an accumulator, and the optimized designs with an accumulator without and with creep; the dotted line is 1%. (b) Probability of a safe stroke and relief chart that keep the over-torque probability at 1% when the stem may be weaker than the data sheet by the factor shown.

The chart basis matters only when stems are not equally strong relative to the operating torque. With the stem strength proportional to the median breakaway torque (Table 1) the two bases are algebraically the same. With allowable torques of 4.0, 3.0, 2.2, 3.5 and 2.0 times the median breakaway torque for the five classes, the best design with a chart from the median breakaway torque has a safe-stroke probability of 56.0% but an over-torque probability of 3.5% for the worst class (butterfly 36 in), while the chart from the allowable torque reaches 60.3% with a worst class at 1.4% (Fig. 12b): the data sheet of the stem is part of the information that the chart needs.

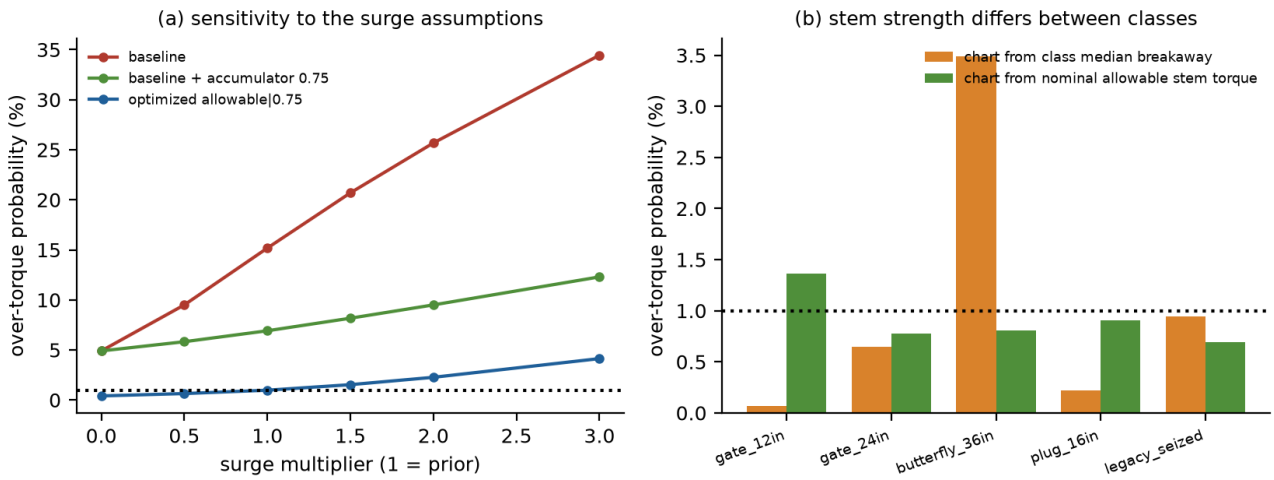


Fig. 12. (a) Over-torque probability against a multiplier of the assumed surge, for the baseline tool, the baseline with an accumulator, and the best design with an accumulator. (b) Over-torque probability of each valve class for the best designs of the two chart bases when the stem strength differs between classes.

6.8 Sensitivity

The torque margin of the tool (the logarithm of its stall capability over the breakaway demand) is controlled by the breakaway torque of the valve (partial rank correlation −0.95; variance-based indices [26] are a natural extension), the supply pressure (0.45) and the gearbox efficiency (0.40); hose length, the hose coefficient, the fouling and the source capacity have correlations below 0.05 in magnitude, so the hose does not limit the torque margin at these flows. The ratio of the peak stall torque to the allowable stem torque is controlled by the allowable stem torque (−0.89), the valve class (−0.60) and the gearbox efficiency (0.49), and it also depends on the hose, through the velocity of the water that the stall has to stop (hose coefficient 0.37) (Fig. 13). A measurement programme should therefore start with the stem strength and the breakaway torque of real valves, and with an instrumented stall test.

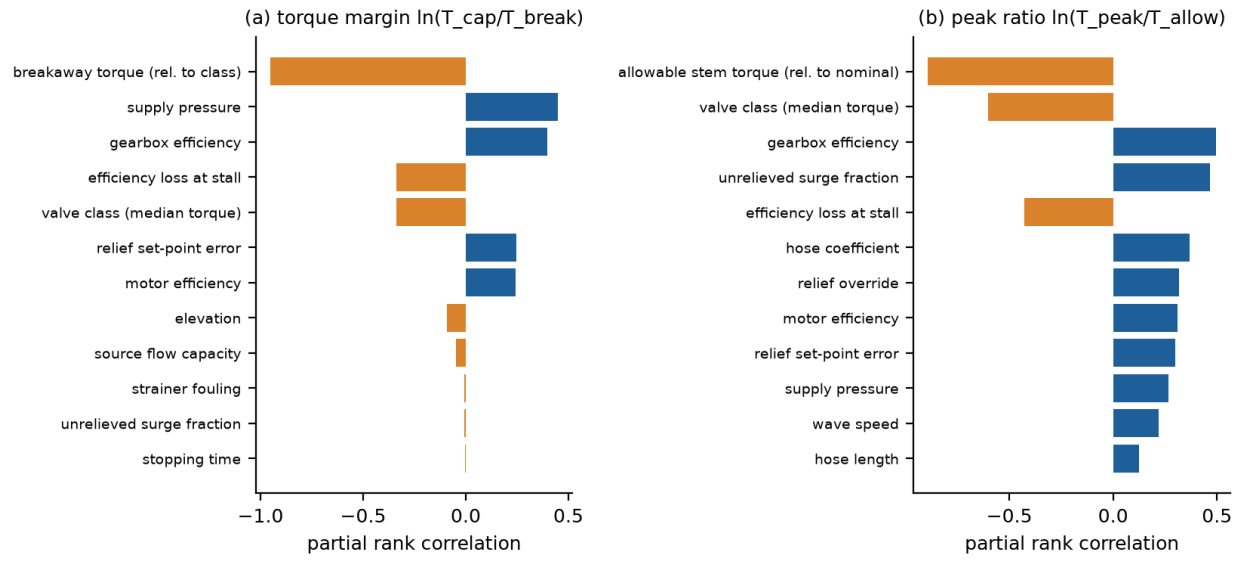


Fig. 13. Partial rank correlation of the inputs with (a) the torque margin ln(T_cap/T_break) and (b) the peak ratio ln(T_peak/T_allow), baseline tool, valve-class mix.

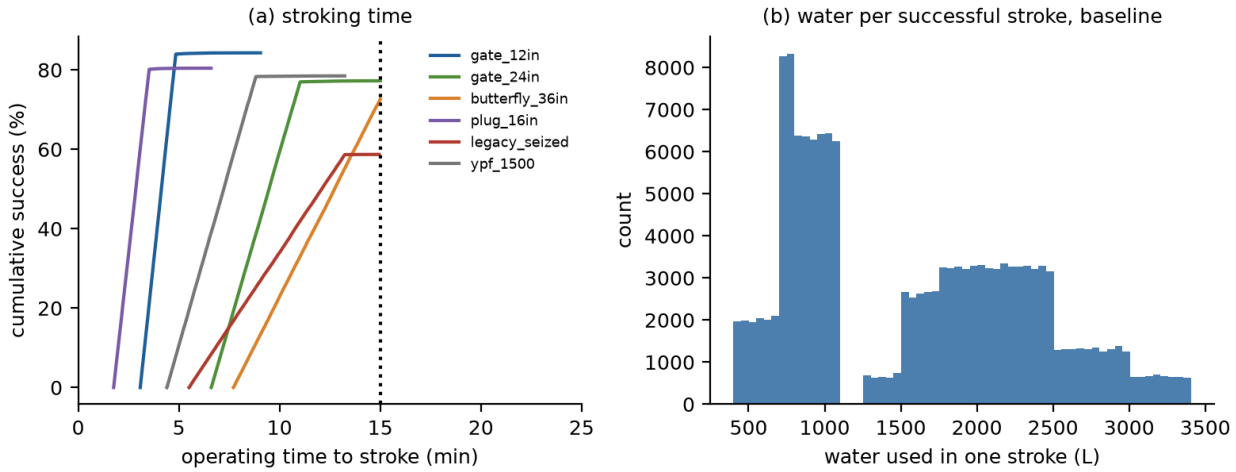


Fig. 14. (a) Distribution of the stroking time of successful strokes by valve class, baseline tool; the dotted line is the 15-minute limit. (b) Water used in one successful stroke.

6.9 Comparison with benchmark tools

Two benchmarks put the tool in context under the same valve draws. A lossless water-hydraulic tool with the same displacement and ratio, no hose or coupling loss, no leakage and no

start-up loss reaches a success probability of 76.4%, against 75.5% for the baseline: the circuit and efficiency losses cost about 0.9 percentage points of success, and the shortfall of the baseline from an ideal tool is therefore not in the hydraulics. Three conventional portable operators (assumed rated torque, power, availability and speed) give: battery-electric operator, 1,000 N.m, 1.5 kW: success 43%, safe 37%; pneumatic operator, 1,500 N.m, 3 kW (plant air): success 33%, safe 21%; engine-driven oil-hydraulic pack, 3,000 N.m, 5 kW: success 85%, safe 43%. The engine-driven pack reaches a higher success probability than the water tool because its rating is higher, not because of any advantage in safety. The conventional operators are capability-limited, with a rated torque that is either too low for the stiff valves or, for the engine-driven pack, without a torque limiter, so that the rated torque would exceed the allowable stem torque in 45% of the draws. These figures rest on assumed ratings and availabilities, not on measured performance; they show where the torque ceiling of each approach lies and do not support a claim of superiority.

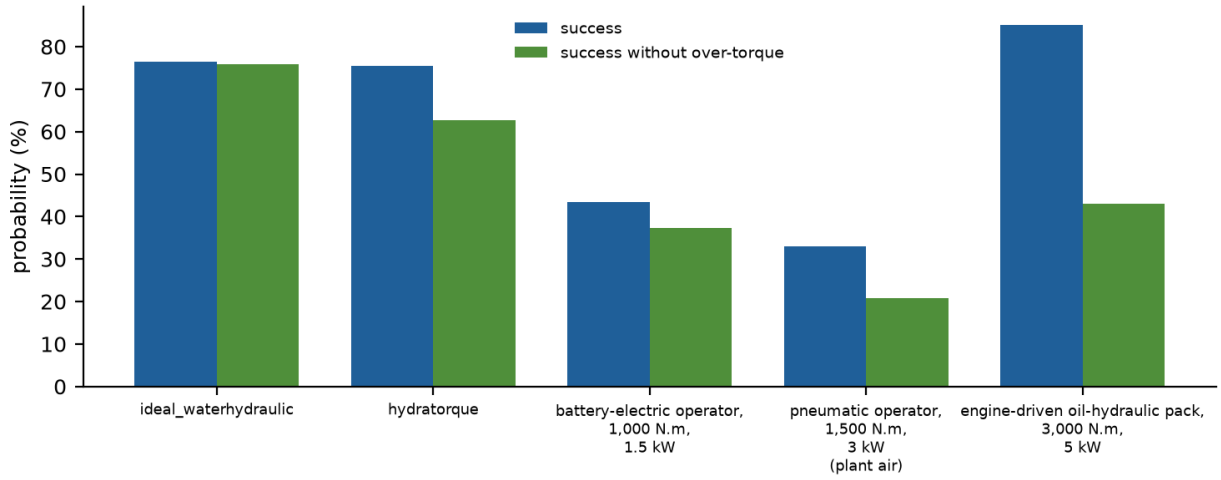


Fig. 15. Success and success without over-torque of the baseline tool, a lossless water-hydraulic tool (no surge model) and three conventional portable operators described by rated torque, power and availability, on the same valve draws.

7. Discussion

7.1 What the study says about the design of the tool

Three results are specific enough to guide a bench programme, each conditional on the assumed priors. First, the safety of the tool is decided at the end of the stroke, not during it: the hose and the circuit do not limit it (a lossless tool gains about one percentage point of success); what limits the safe-stroke probability is the torque that the tool delivers at stall, set by a relief valve whose pressure is low for small valves and which cannot relieve a surge in time. The relief chart and the surge, not the motor, are the design problem. Second, a chart computed from the stem strength, with a hose-side accumulator, brings the over-torque probability to 1% at a cost of 3 to 9 percentage points of safe strokes relative to the baseline; a slow approach to the end of travel recovers 4 to 6 points of it, and the rest is paid in valves that the tool then cannot break free, which need a higher-pressure source or another tool. Third, the water budget is a design variable: a stroke uses of the order of a thousand litres, and whether that is acceptable depends on a spare flow that the operator has to know.

The recommended starting configuration for the bench programme is the speed-constrained design with an accumulator and creep: a 473 cc motor with a 109:1 reduction (motor speed 534 rpm at the flow limit), a flow limit of 67 gpm with creep flows of 52 gpm at the start and 32 gpm at the end of travel, and a relief chart at 0.52 of the allowable stem torque. On independent draws it gives a safe stroke in 63.4% of the draws with an over-torque probability of 0.98% (upper 95% limit 1.02%); under the strict compliance rule the best speed-constrained design gives 62.3% with an upper limit of 0.74%. These figures hold for verified stem strengths and a relief valve that responds within about 0.1 s (Sections 4.3 and 6.7).

The results suggest the following design hypotheses, to be tested on the bench before they are rules: a relief chart set at about half of the nominal allowable stem torque of the valve; a hose-side accumulator or an equivalent surge absorber; a variable flow limiter that creeps for the first turn and the last two turns of the stroke; a flow limit between 29 and 85 gpm for strokes of 30 to 70 turns, with the adaptive reduction on shared sources; a product of displacement and gear ratio of about 30,000 to 51,500; and a dedicated source for the large valves and for any site that has not measured its spare firewater flow. The numbers will move with the priors, and the robustness analysis shows in which direction: a weaker stem or a larger surge calls for a lower chart.

7.2 Relation to existing devices and what is claimed

Existing portable valve operators run on electricity, batteries, gasoline or compressed air; hydrant operating apparatus carries a hose coupler and a drive socket for the hydrant stem [11], a hydrant-attached remote-controlled valve opens a flow channel with a pilot-operated piston [12], and an oscillatory actuator controls a valve on a fire truck [13]. Water-hydraulic motors [5,6] and water hammer [16] are established. The contribution claimed here is neither of these but the joint probabilistic treatment of a firewater-powered tool: the safe stroke as a combined event, the surge-aware relief chart, the water and firefighting-reserve constraints, and the test of the optimum against wrong priors. The comparison with conventional operators (Section 6.9) rests on assumed ratings, no claim of superiority is made, and a full prior-art and freedom-to-operate review would be needed before any claim of novelty.

7.3 Deployment constraints

Four constraints follow from the setting and are not captured by the numbers above. The tool must not compromise the firefighting reserve; the model turns this into an operating rule that is only as good as the knowledge of the spare flow. The reaction torque must be carried by an

anchor, whose failure is a hazard to the operator that the model does not include (Section 7.4). Water-filled equipment freezes, so use in cold regions, such as the Austral Zone of Patagonia named in the industrial challenge [3], needs drain-down procedures or an anti-freeze flush, and a mechanical tool of this kind would have to be assessed as non-electrical equipment for explosive atmospheres [15] and under the risk-assessment principles for machinery [27] before use in a classified area; the valve is subject to its own pressure-class rules [28]. Finally, the technology readiness of the concept is that of a computational study: the industrial challenge asks for a solution that is commercially available or at a readiness level of 5 to 9 with proven field reliability [3], which this concept does not meet.

7.4 Preliminary mechanical feasibility

The operating envelope of Section 6.6 is only useful if the loads it implies are mechanically attainable in a portable tool. Table 10 gives screening estimates for the draws of the model: the stall torque that the tool delivers, the output torque rating that a planetary gearbox would need (1.5 times the 99.9th percentile of the peak stall torque), the reaction force and the section modulus of a steel reaction arm (yield 355 MPa, safety factor 2.0, arm length 0.6 m), the motor speed and hydraulic power, the velocity in the hose and the gas volume that an isothermal accumulator would need to absorb 75% of the kinetic energy of the water column. These are screening estimates of required quantities, not selected components: the arm is a cantilever of one load case, the gearbox rating ignores shock loads and duty, and the accumulator volume is a required gas volume that ignores the compliance of the hose and the precharge of a real unit; it is not a commercial size. The dimensions and mass of the assembly cannot be estimated without component data; the loads define the data to ask of the manufacturers.

Quantity	Baseline	Best, no accumulator	Best, accumulator and creep	No speed limit
Peak stall torque, median (N.m)	2,353	1,547	1,719	1,647

Quantity	Baseline	Best, no accumulator	Best, accumulator and creep	No speed limit
Peak stall torque, 99th percentile (N.m)	4,982	3,347	3,690	3,676
Gearbox output rating, 1.5 x 99.9th percentile (N.m)	8,543	5,717	6,195	6,177
Reaction force at the arm tip, 99th percentile (kN)	8.3	5.6	6.2	6.1
Arm section modulus, 99th percentile (cm3)	28	19	21	21
Motor speed at the flow limit (rpm)	1,136	414	534	4,687
Hydraulic power at the motor, 95th percentile (kW)	1.3	1.1	1.4	2.2
Velocity in the hose, 95th percentile (m/s)	2.4	1.4	2.7	3.9
Required accumulator gas volume, median (L)	(0.8)	(0.2)	1.3	1.9
Required accumulator gas volume, 95th percentile (L)	(4.1)	(1.0)	6.0	9.5
Required accumulator gas volume, 99th percentile (L)	(6.7)	(1.6)	10.0	16.0

Table 10. Preliminary mechanical loads for the baseline tool (D = 200 cc, R = 250), the best speed-constrained design without an accumulator or creep (D = 305 cc, R = 100) and the best speed-constrained design with an accumulator and creep (D = 473 cc, R = 109) and, for comparison, the best design without the speed limit (D = 77 cc, R = 554); 100,000 draws, valve-class mix. Accumulator volumes are screening estimates of the gas volume required to absorb 75% of the surge energy, not selected commercial sizes; values in parentheses are hypothetical, for designs that have no accumulator.

Four observations follow. The stall torque is of the order of 3.3 to 5.0 kN.m at the 99th percentile, so the gearbox must be rated for 5,717 to 8,543 N.m at the output and the reaction arm must carry 5.6 to 8.3 kN at 0.6 m, a modest steel section (a section modulus of 19 to 28 cm3); the optimized designs lower these loads by 26% to 33% because their relief setting is lower. The required accumulator gas volume of the recommended design is 1.3 L at the median and 6.0 L at the 95th percentile (hypothetically 4.1 L at the 95th percentile for the baseline). The motor speed is part of the search: the recommended design runs a 473 cc motor at 534 rpm at the flow limit; without the 3,000 rpm limit the best design would run a 77 cc motor at 4,687 rpm and reach a safe-stroke probability of 63.7% out of sample, against 63.4% for the speed-constrained

design, so the limit costs 0.4 percentage points. A limit of 1,500 rpm lowers the best in-sample probability of a safe stroke of this family to 63.2%. Finally, the velocity of the water in the hose reaches 2.7 m/s at the 95th percentile for the recommended design, which would favor the larger hose sizes.

8. Limitations and future validation

The principal limitation is that every input distribution is an assumed prior: no published distribution of breakaway torques of the valves of interest was found, and the surge parameters, the efficiency at stall and the spare flows of firewater sources are assumptions. A larger number of simulations does not compensate for uncertain priors; Sections 4.4 and 6.7 are the only protection. The model is lumped: the relief valve acts as an override with sampled overshoot, the hose compliance is a sampled wave speed, and the stem is a single strength with a sampled yield margin. The inputs are independent although valve size, age, stem strength and breakaway torque are correlated. The surge model is a screening approximation, checked against classical results and a transient simulation of its own assumptions (Section 4.3) but not against a measured stall transient; the mechanical loads of Section 7.4 are screening estimates, and the 3,000 rpm limit on motor speed is an assumed requirement. The model is verified numerically, not validated physically [22]; its numbers should not be quoted as field reliability.

A validation programme that would replace the assumed priors has six steps. (i) A survey of breakaway, running and allowable stem torques of the valves to be served (2 to 16 inches) calibrates the valve classes and the stem strengths. (ii) A bench rig with a torque transducer measures motor torque and leakage against pressure drop and speed, including the loss at start-up. (iii) An instrumented stall test with fast pressure transducers measures the surge, the wave speed and the response of the relief valve, with and without an accumulator and creep. (iv)

Hydrant and firewater flow tests [9] give the spare flows of shared sources. (v) A field pilot with a hazard identification tests the operating procedure. (vi) A hazardous-area assessment closes the certification path.

9. Conclusions

1. A simulation of a firewater-powered portable valve operator, with an explicit hydraulic circuit and the surge on stall, shows that a stroke without exceeding the allowable stem torque occurs in 63% of the draws for the baseline tool, against the 80% success reported by the first-order model; the over-torque probability is 15%, not 5%, and the stem-damage probability is 1.1%. These are simulation outputs under assumed priors.

2. The surge on stall is the main source of over-torque, and it is largest in relative terms for small valves, whose relief pressure is low. A hose-side accumulator lowers the over-torque probability of the baseline tool from 15% to 7%, and a slow creep flow for the last two turns of the stroke lowers it from 15% to 9% (two turns at 20 gpm), at a cost of about 0.8 minutes of stroking time.

3. Under a 1% limit on the over-torque probability, which is an illustrative constraint, a design exists in every family searched, and it holds out of sample; the cost is 3 to 9 percentage points of the safe-stroke probability. A single configuration is almost as good as valve-class-specific ones; the relief chart is what has to be specific.

4. A stroke uses about 1,600 L of water at the 60 gpm limit; the 50 to 100 L/min of the first-order study is the flow for 1 to 2 rpm, too slow for a 15-minute stroke. The firefighting reserve is protected only if the spare flow of a shared source is known, and a dedicated source is the safe default for large valves.

5. The surge relations of the model reproduce the classical Joukowsky and Michaud results and are conservative in 97% of random transient cases, but only for a relief valve that responds

within about 0.1 s; they are verified, not validated, and an instrumented stall test is the experiment that the model needs.

6. The optimized configuration cannot be considered operationally safe without verified stem-strength information: a stem 10% weaker than the data sheet raises the over-torque probability to 2.4%, and 30% weaker to 14%; the over-torque probability of a design chosen at an estimated 1% is 0.79% to 0.99% with upper 95% limits up to 1.03%, and a strict compliance rule costs 0.0 to 4.2 percentage points of safe strokes.

7. The chart basis matters only if stem strength differs between valve classes relative to their operating torque, in which case a chart from the allowable stem torque is required.

Declarations

Funding. The work received no external funding. It was carried out by the author as an independent researcher, with no grant, contract or award; the industrial challenge that motivated it states no prize, and the concept was prepared at the author's own expense.

Competing interests. The author has no financial or non-financial interests that are directly or indirectly related to the work. The concept was prepared by the author as a white paper for an industrial scouting challenge [3]; no patent has been filed and no contract or payment has resulted.

Availability of data and material. All results are generated by the code archived at https://doi.org/10.5281/zenodo.23969346 (all versions) and available at https://github.com/sandlerleon/hydratorque-firewater-valve-actuation-mc; the manuscript and its Supplementary Information are archived at https://doi.org/10.5281/zenodo.23969344 (all versions). No measured data were used.

Code availability. The simulation (HydraTorque_MC.py, version 2.2.0, numpy and matplotlib only), its automated tests and the figure and manuscript builders are in the repository; python HydraTorque_MC.py --reproduce regenerates every number and figure (seed 2026); a fresh clone of the repository reproduced every number and figure of the paper identically (same machine, Python 3.11, numpy 2.4.6).

Author contributions. The author is the sole contributor.

Use of generative AI. During the preparation of this work the author used a generative AI tool (Claude, Anthropic) for code development (the extended hydraulic model, the optimization and the automated tests), for the analysis scripts and for drafting and editing the text. After using the tool, the author reviewed and edited the content and takes full responsibility for the content of the publication.

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